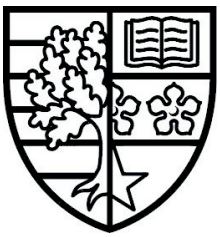
# Skin Lesion Classification on FPGA

*by* Lewis Turner

Research Report

*Submitted in partial fulfilment for the degree of*
BSc (Hons) Computer Science

*Supervised by* Dr. Rob Stewart, Ms Tess Watt



Heriot-Watt University
School of Mathematical and Computer Sciences

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| **Type of assessment** | Individual |
| **Document title** | Skin Lesion Classification on FPGA |
| **Student name** | Lewis Turner |
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• Google Gemini 3.8 Flash (Extended Thinking) to assist in searching for relevant research papers.

• Google Gemini 4.0 Argon (thinking: Low - High) to assist setting up FPGA for use over Wifi.

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## Acknowledgements

Acknowledgements text.

## Abstract

Abstract text.

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# 1 Introduction

Summary of project motivation & context

## 1.1 Background and Motivation

- Using papers (mainly) from literature review to provide background context
- Motivations for doing this project
- Identifying problems to be solved to achieve objectives

## 1.2 Aim and Objectives

State project aims
Table of objectives with IDs to reference later

# 2 Literature Review

Grouping topic summaries, critical assessment, gaps
Possibly use RADAR?

## 2.1 Search Methods

I used Google Gemini 3.8 Flash on thinking mode High (October 2026 version) to source the most recent academic papers on a particular topic. I would ask it to create a table consisting of the papers: title, year, summary, DOI or similar URL. This allowed me to leverage the LLM's speed for searching the internet like a more powerful keyword search.

I occasionally used Google Scholar as well to search a specific paper title provided by Gemini to find the original source PDF/website.

## 2.2 Skin Lesion Classification Models

Papers on recent skin classification models

## 2.3 Machine Learning Models on FPGA

Papers more generally on methodology for putting ML models on FPGA

## 2.4 Compressing Machine Learning Models to FPGA

Papers on useful compression techniques for certain model types

## 2.5 Further Research Recommendations

Identify gaps, need to repeat any research

## 2.6 Acknowledgements

Sources for literature/statements
Example citation to stop LaTeX complaining: [1].

# 3 Methodology

Detailed description of necessary methods, requirements, research questions that will be employed/ needed to complete the project

## 3.1 Requirements

MOSCOW
Enumerated to be easier to reference again

## 3.2 Plan for Execution

Explain how I plan to use experiments to fulfil all the requirements & objectives

## 3.3 Experiment Planning

How I plan to conduct experiment(s) and what research question does it address/answer (detailed)

# 4 Feasibility

## 4.1 Preliminary Work

Summary of work carried out to prepare for this project to validate feasibility.

## 4.2 Evaluation Strategy

Identify for the project:
- What needs evaluating
- Why it should be evaluated
- How it should be evaluated

# 5 Conclusions

Short summary of this doc

## References

[1] Alan M. Turing. 2009. Computing Machinery and Intelligence. In *Parsing the Turing Test: Philosophical and Methodological Issues in the Quest for the Thinking Computer*, Robert Epstein, Gary Roberts, and Grace Beber (Eds.). Springer Netherlands, Dordrecht, 23–65. doi:10.1007/978-1-4020-6710-5_3

# 6 Project Management

## 6.1 Gantt Chart

Plan of activities to be carried out in semester 1, then future plan for semester 2

## 6.2 Analysis of Risks

Including mitigation plans

## 6.3 PLES

A well-researched consideration of any Professional, Legal, Ethical, and Social Issues pertinent to the project. (e.g. codes of conduct (BCS), codes of practice, standards, computer law, ethical decision making, intellectual property, social aspects, copyright, data protection, and so on).

# 7 Gen AI Summary

Summarising use of GenAI in project/deliverables